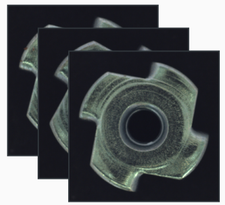
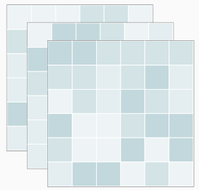
(a) Fixed before any query (zero training)

NORMAL IMAGES (C1)



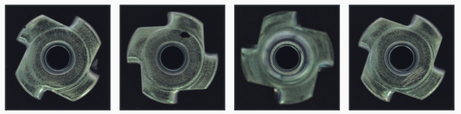
5–20 unlabeled normal images

frozen CLIP, layer-18 patch tokens



Normal patch bank *B*
M × 576 patches

SUPPORT SET *S*



bent color flip scratch

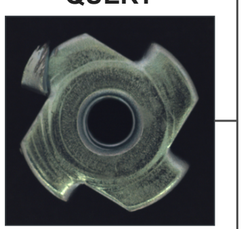
K images per defect type, image-level labels

same path as the query

Class prototypes p_c
centred class means of g, h_A, h_R

(b) For each query, independently

QUERY



whole image only

CLIP ViT-L/14@336
frozen · layer 18

patch tokens *t* (24 × 24)

embedding

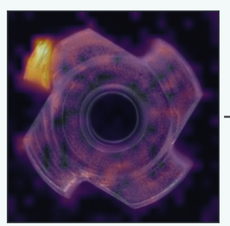
DINOv2 ViT-S/14
frozen

$g(x)$

global feature: CLIP embedding ⊕ DINOv2 patch mean, each L2-normalised

Normal-referenced evidence selection (C1)

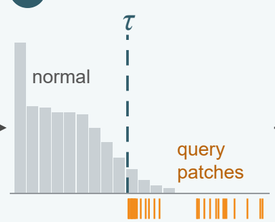
1 Score



$a(t) = 1 - \max_{b \in B} \cos(t, b)$

distance to the nearest normal patch

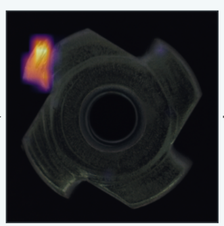
2 Gate



$\tau = Q_{0.99}$ of leave-one-out normal-patch scores in *B*

computed once; normal variation ends at τ

3 Weight



$w(t) = \max(a(t) - \tau, 0)$

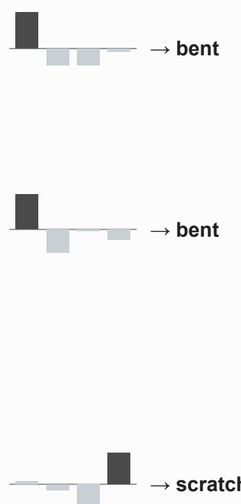
excess over τ; 32 of 576 patches exceed τ

same *w*

Appearance h_A
$\Sigma w\, t \,/\, \Sigma w$

Deviation h_R
$\Sigma w(t - \mathrm{nn}_B(t)) \,/\, \Sigma w$

SimpleShot
vs prototypes, z-scored



Fused score
sum of the three



bent color flip scratch

ŷ = bent

Heat maps, τ, patch scores and score bars: computed for this query (MVTec-FS metal_nut, K = 5, seed 111); grids and vectors are schematic.